Project – III

1 Constraint Model

We adopt the natural mixed-integer formulation of capacitated facility location with vehicle workload, and then relax all integrality so that every decision variable becomes continuous on a real interval. The resulting LP is what `src/lpinstance.py` gives to OR-Tools.

We use an aggregated form: v_f counts vehicles at facility f as a single continuous quantity, rather than indexing each vehicle slot separately. This keeps the model compact (one variable per facility for vehicles, one per customer-facility pair for assignment) at the cost of some LP tightness: distance is constrained in aggregate, not per truck. The trade-off is acceptable because LP only needs to certify a lower bound, not implement a plan.

2 Decision Variables

All variables are continuous (no integrality).

Let C and F denote the customer and facility index sets:

- $y_f \in [0, 1]$, one per facility $f \in F$: the fraction to which facility f is opened. In IP, this is binary. In LP, it can be any real value in $[0, 1]$.
- $x_{c,f} \in [0, 1]$, one per pair $(c, f) \in C \times F$: the fraction of customer c's demand that is served from facility f. In IP, this is binary (each customer is assigned to exactly one facility). However, LP allows split assignments.
- $v_f \in [0, V]$, one per facility $f \in F$: the continuous number of vehicles deployed at facility f. In IP, this is an integer value.

3 Constraints

1. Each customer is fully served across some combination of facilities:

$$\sum_{f \in F} x_{c,f} \;=\; 1 \qquad \forall c \in C.$$

2. A customer may only draw service from a facility to the extent that facility is open:

$$x_{c,f} \;\le\; y_f \qquad \forall c \in C,\; f \in F.$$

(In IP, it would have to be $x = 1 \Rightarrow y = 1$.)

3. The demand served from a facility f may not exceed its capacity, scaled by its openness. (A half-open facility carries only half the capacity.)

$$\sum_{c \in C} d_c\, x_{c,f} \;\le\; q_f\, y_f \qquad \forall f \in F.$$

4. Vehicle workload: Total round-trip distance dispatched from a facility f cannot exceed the workload that v_f vehicles can absorb:

$$\sum_{c\in C} \delta_{c,f}\, x_{c,f} \;\leq\; T\, v_f \qquad \forall f \in F.$$

   We have this instead of a per-vehicle limit. With v_f being continuous, we can split distance across fractional vehicles.

5. A closed facility cannot deploy vehicles, and we cannot deploy more than V:

$$v_f \;\leq\; V \cdot y_f \qquad \forall f \in F.$$

Objective. Minimise the sum of opening, service, and truck-usage costs:

$$\min \sum_{f\in F} o_f\, y_f \;+\; \sum_{c\in C}\sum_{f\in F} a_{c,f}\, x_{c,f} \;+\; u \sum_{f\in F} v_f.$$

4 Effort

Approximately 20 hours, including modeling, debugging, and writing this report.